Shivam Ajay Saraswat

+91 9503611319 shivamsaraswat551@gmail.com
github.com/WILLSHAPER5-10-5 linkedin.com/in/shivam-saraswat-705218238
SC Code: SC25M145 ◇ Manufacturing Technology ◇ Aerospace Engineering



Education

Indian Institute of Space Science & Technology (IIST) **2025 – 2027**
M.Tech in Manufacturing Technology — CGPA: 6.03 *Thiruvananthapuram, Kerala*

MIT-WPU, Pune **2019 – 2023**
B.Tech in Mechanical Engineering — CGPA: 8.68 *Pune, Maharashtra*

Experience

Vikram Sarabhai Space Centre (VSSC) **2026**
Intern - Mathematical and Analytical Modelling of Metal 3D Printed Compliant Mechanism

Advanced Manufacturing Technology Development Centre (AMTDC) **2026**
Intern - Development of Dashboard for CNC Machines

- Designed and developed dashboard for CNC machines (Jyoti MTX 300)
- Used Eclipse Milo for server generation
- Produced technical documentation and reports

Armament Research and Development Establishment (DRDO) **2023**
Design Intern - Servo Mechanism for Auto Cannon

- Designed and analyzed servo mechanism for gun laying system of 2A42 30mm auto cannon
- Conducted mechanical simulations, stress analysis, and kinematic studies using CAE tools (ANSYS, Simulink)
- Produced comprehensive technical documentation and design reports for military-grade applications

Projects

Design and Analysis of a Random Positioning Machine to Simulate Microgravity **Present**

- Aim to develop a Random Positioning Machine (RPM) designed to simulate microgravity environments through multi-axis rotational kinematics.
- Aim to execute rigorous structural analysis and FEA simulations to validate load-bearing capacity, ensuring high factors of safety under dynamic loading conditions.
- Aim to optimize frame geometry and component selection for structural rigidity, ease of fabrication, and ergonomic user interaction.

Design and Analysis of Modular Motorised Wheelchair **2022**

- Engineered a modular wheelchair architecture integrating a motorized propulsion system, focusing on scalable mechanical interfaces for enhanced accessibility.
- Conducted multi-scenario FEA simulations to analyze stress distribution and deformation, ensuring structural integrity under peak dynamic loads and diverse terrain impacts.
- Performed topology optimization and iterative design cycles to balance ergonomic requirements with mass reduction and manufacturing feasibility.

Design and Fabrication of Line Following Robot **2022**

- Developed an autonomous mobile platform utilizing sensor-fusion and path-tracking algorithms precision navigation.
- Engineered a mechanical chassis, employing structural modeling to optimize weight distribution and ensure component stability under operational vibrations.
- Synthesized obstacle detection and automated decision-making logic, integrating an Arduino-based control system for real-time response and kinematic execution.

Design of Small Scale Indoor Hydroponic System (IoT-Controlled) **2022**

- Developed an automated hydroponic architecture utilizing IoT-based controls to maintain high-fidelity environmental parameters for plant growth.
- Integrated a multi-sensor array for real-time data acquisition of pH, fluid levels, and hydrothermal conditions, ensuring precise feedback loops for system stability.
- Synthesized a centralized mobile interface for remote monitoring and automated management, optimizing resource consumption through data-driven decision-making.

Technical Skills

CAD Software: Creo, Fusion 360, Solidworks, Inventor, OnShape, CATIA, FreeCad, AutoCAD

CAE Software: ANSYS Workbench, Abaqus

Programming Languages: Java, Python, C, C++, MATLAB, Simulink

3D Printing: Ultimaker Cura, Prusa Slicer, SimuFact Additive

Core Competencies: Mechanical Design, FEA & Structural Analysis, Servo Systems, Manufacturing Processes, Robotics

Languages: English, Hindi, Marathi

Leadership & Responsibilities

- **Workshop Instructor, SKETCH Club** – Conducted CAD/CAE workshops for 50+ students, focusing on parametric modeling.
- **Member, Workshop & Teaching Team** – Developed curriculum for design education initiatives, emphasizing computational analysis workflows.

Certifications & Achievements

- **DSC WOW Hackathon** – Top 20 Finalists
- **National Engineering Olympiad** – Round 1 Qualified
- **MIT-WPU Workthon** – 2nd Prize Winner
- **Design, Technology and Innovation** – NPTEL Certification
- **Wind Energy** – Coursera Certification
- **HTML, JAVA, CSS** – Coursera Certification

References

- **Dr. V S Sooraj** – Professor, Department of Aerospace Engineering, IIST — sooraj@iist.ac.in
- **Dr. Nilesh Dhokane** – Deputy Registrar, MIT-WPU — nilesh.dhokane@mitwpu.edu.in
- **Dr. Pankaj Dhatrak** – Associate Professor, MIT-WPU — pankaj.dhatrak@mitwpu.edu.in
- **Mr. Arun Mali** – Assistant Professor, MIT-WPU — arun.mali@mitwpu.edu.in
- **Dr. Ashish Utage** – Assistant Professor, MIT-WPU — ashish.utage@mitwpu.edu.in